

Information and Estimation Theory Lecture 04: Entropy rates of discrete stochastic processes

A.Y. 2025/26 (TIS2526)
Giuseppe Baruffa (Mod. IT)
Computer Engineering and Robotics MD

A.D. 1308 unipg
DIPARTIMENTO DI INGEGNERIA
Università degli Studi di Perugia
DSP Laboratory
MCMXCVIII

Stochastic Processes

- AEP says that $nH(X)$ bits suffice on the avg. to describe n i.i.d. RVs
  - But what if the RVs are dependent? Is the PMF dependent on n?

- We will show that $H(X_1, X_2, \ldots, X_n)$ grows (asymptotically) linearly with n at a rate $H(X)$ called the **entropy rate** of the process

- A **stochastic or random process (SP)** is an indexed sequence of RVs $\{X_i : i \in \mathbb{Z}\}$
  - In general, there can be an arbitrary dependence among the RVs

- The process is characterized by the joint PMFs

$$\Pr\{(X_1,X_2,\ldots,X_n) = (x_1,x_2,\ldots,x_n)\} = p(x_1,x_2,\ldots,x_n), \quad (x_1,x_2,\ldots,x_n) \in \mathcal{X}^n$$

for $n = 1,2,\ldots$

Example: Random Walk

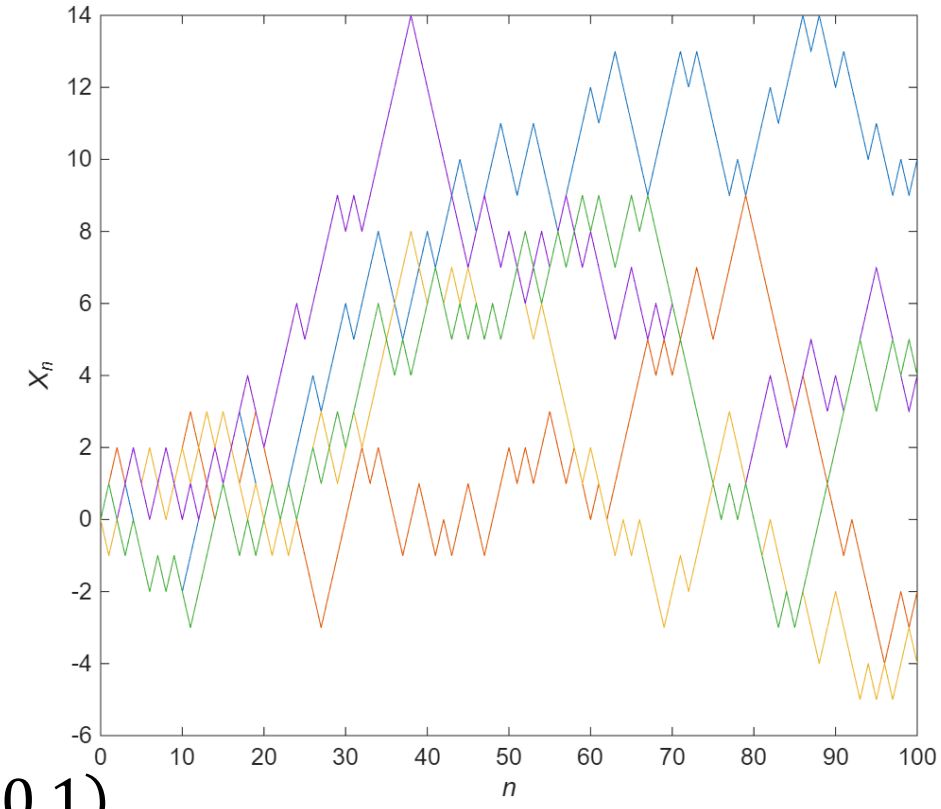
- Let $Y_1, Y_2, \ldots$ be i.i.d. RVs and $Y_i = \pm 1$ with equal probability
- Let $X_0 = 0$ and

$$X_k = Y_1 + \ldots + Y_k, \qquad \text{for all } k \geq 1$$

- This gives a PMF over the sequences $X_0, X_1, \ldots$, and thus defines a discrete time SP
- This process is known as the simple 1D **random walk**
  - X_0: $x_0 \in \{0\}$
  - X_1: $x_1 \in \{-1,+1\}$
  - X_2: $x_2 \in \{-2,0,+2\}$
  - X_3: $x_3 \in \{-3,-1,+1,+3\}$
  - X_4: $x_4 \in \{-4,-2,0,+2,+4\}$ etc.

$$|\mathcal{X}_n| = n + 1$$

- By the central limit theorem, for large n, the PMF of $(1/\sqrt{n})X_n$ converges to pdf $\mathcal{N}(0,1)$



Stationarity

- A SP is **stationary** if the joint PMF of any subset of the sequence of RVs is invariant with respect to shifts in the (time) index

$$\Pr\{X_1 = x_1, X_2 = x_2, \dots, X_n = x_n\} = \Pr\{X_{1+l} = x_1, X_{2+l} = x_2, \dots, X_{n+l} = x_n\}$$

for every *n* and every shift *l* and for all $x_1, x_2, \dots, x_n \in \mathcal{X}$

- The random walk is not stationary, indeed for *n*=2 and *l*=1,

$$\Pr\{X_1 = x_1, X_2 = x_2\} \neq \Pr\{X_2 = x_1, X_3 = x_2\}$$

- However, differencing the random walk gives a stationary SP

$$Z_n = X_n - X_{n-1}$$

$$Z_n : z_n \in \{-1, +1\}$$

Markov Chains

- A discrete SP $X_1, X_2, \ldots$ is a **Markov chain** (discrete RV) or a **Markov process** (cont. RV) if, for $n=1,2,\ldots$, for all $x_1, x_2, \ldots, x_n, x_{n+1} \in \mathcal{X}$,

$$\begin{aligned}\Pr\{X_{n+1} = x_{n+1} | X_n = x_n, X_{n-1} = x_{n-1}, \ldots, X_1 = x_1\} \\ = \Pr\{X_{n+1} = x_{n+1} | X_n = x_n\}\end{aligned}$$

- The joint distribution of the RVs can be written as

$$\begin{aligned}p(x_1, x_2, \ldots, x_n) &= p(x_1)p(x_2|x_1)p(x_3|x_1, x_2)\cdots p(x_n|x_1, x_2, \ldots, x_{n-1}) \\ &= p(x_1)p(x_2|x_1)p(x_3|x_2)\cdots p(x_n|x_{n-1})\end{aligned}$$

- The Markov chain is **time invariant (TI)** if the conditional distribution $p(x_{n+1}|x_n)$ does not depend on n; for $n=1,2,\ldots$,

$$\Pr\{X_{n+1} = b | X_n = a\} = \Pr\{X_2 = b | X_1 = a\} \qquad \text{for all } a, b \in \mathcal{X}$$

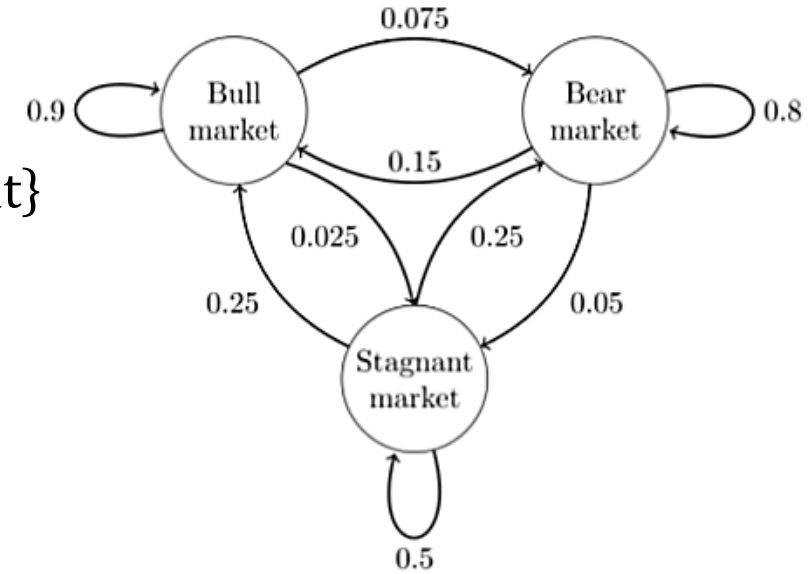
  - The Markov chain is TI unless otherwise stated

Stochastic Matrix

- If $\{X_i\}$ is a Markov chain, X_n is called the **state** at time n
  - I.e., the state is the value of the RV at time n
- A TI Markov chain has an initial state and a **probability transition matrix P =** $[P_{ij}]$, $i,j \in \{1,2,\ldots,m\}$, where $P_{ij} = \Pr\{X_{n+1} = j \mid X_n = i\}$
  - Also called **stochastic matrix**

- Example: weekly market trend
  - State is one of $\mathcal{X} = \{\text{Bull}, \text{Bear}, \text{Stagnant}\}$
  - Stochastic matrix is

From/To	Bull	Bear	Stagnant
Bull	0.9	0.075	0.025
Bear	0.15	0.8	0.05
Stagnant	0.25	0.25	0.5



Markov chain state transition graph

Chain State Probabilities

- If the PMF of the RV at time n is $p(x_n)$, the PMF at time n+1 is

$$p(x_{n+1}) = \sum_{x_n \in \mathcal{X}} p(x_n) P_{x_n x_{n+1}}$$

- Let $\mathbf{p}_n = [p_n(i)]$ the probability that chain is in state i at time n
- Time advancing by 1 step is a vector-matrix multiplication

$$\mathbf{p}_{n+1} = \mathbf{p}_n \mathbf{P}$$

- Time advancing by k steps is simply obtained as

$$\mathbf{p}_{n+k} = \mathbf{p}_{n+k-1}\mathbf{P} = \mathbf{p}_{n+k-2}\mathbf{P}\mathbf{P} = \cdots = \mathbf{p}_n \mathbf{P}^k$$

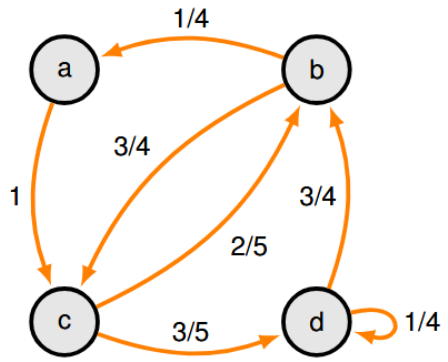
- Exponentiation of the stochastic matrix gives the probability of transition after k steps

$$\mathbf{P}^k = \left[P_{ij}^k\right] = \Pr\{X_{n+k} = j | X_n = i\}$$

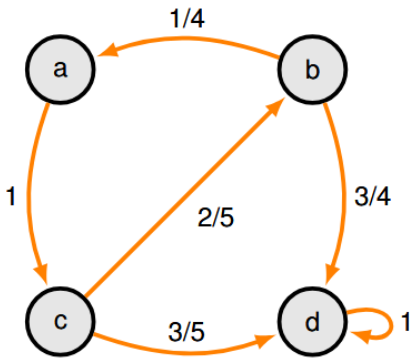
Reducibility and Periodicity

- If you can go with probability higher than zero from any state *i* of the Markov chain to any other state *j* in a finite number *m* of steps, it is **irreducible**

$$\exists m \geq 0 : \left[P_{ij}^{m}\right] > 0, \forall ij$$



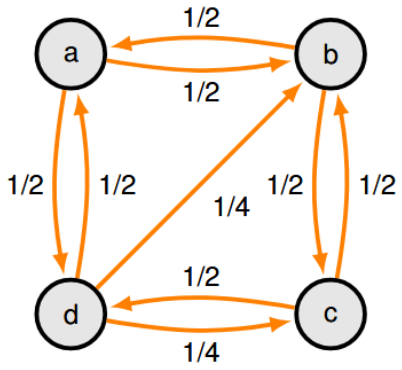
✓ irreducible



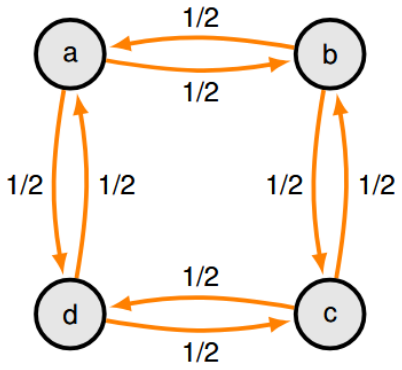
× not-irreducible (thus reducible)

- If the greatest common divisor of the lengths of different paths from a state to itself is 1, it is **aperiodic**

$$\gcd\left\{k : k > 0, \left[P_{ii}^{k}\right] > 0\right\} = 1 \quad \forall i$$



✓ Aperiodic

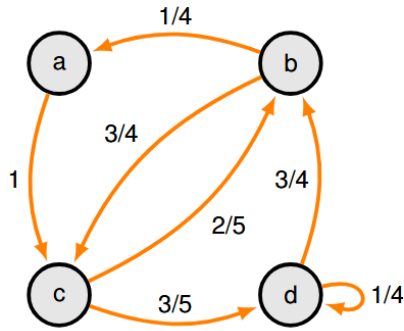


× Periodic

Example: irreducible

```
P =
         0         0    1.0000         0
    0.2500         0    0.7500         0
         0    0.4000         0    0.6000
         0    0.7500         0    0.2500
```

```
P^0 =
     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1
```




```
P^1 =
         0         0    1.0000         0
    0.2500         0    0.7500         0
         0    0.4000         0    0.6000
         0    0.7500         0    0.2500
```

```
P^2 =
         0    0.4000         0    0.6000
         0    0.3000    0.2500    0.4500
    0.1000    0.4500    0.3000    0.1500
    0.1875    0.1875    0.5625    0.0625
```

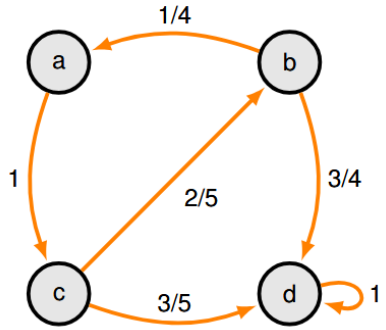
```
P^3 =
    0.1000    0.4500    0.3000    0.1500
    0.0750    0.4375    0.2250    0.2625
    0.1125    0.2325    0.4375    0.2175
    0.0469    0.2719    0.3281    0.3531
```

```
P^4 =
    0.1125    0.2325    0.4375    0.2175
    0.1094    0.2869    0.4031    0.2006
    0.0581    0.3381    0.2869    0.3169
    0.0680    0.3961    0.2508    0.2852
```

```
P^8 =
    0.0783    0.3269    0.3230    0.2718
    0.0807    0.3235    0.3295    0.2663
    0.0817    0.3330    0.3235    0.2617
    0.0849    0.3272    0.3329    0.2550
```

```
P^32 =
    0.0820    0.3279    0.3279    0.2623
    0.0820    0.3279    0.3279    0.2623
    0.0820    0.3279    0.3279    0.2623
    0.0820    0.3279    0.3279    0.2623
```

Example: reducible




```
P =
         0         0    1.0000         0
    0.2500         0         0    0.7500
         0    0.4000         0    0.6000
         0         0         0    1.0000
```

```
P^0 =
     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1
```

```
P^1 =
         0         0    1.0000         0
    0.2500         0         0    0.7500
         0    0.4000         0    0.6000
         0         0         0    1.0000
```

```
P^2 =
         0    0.4000         0    0.6000
         0         0    0.2500    0.7500
    0.1000         0         0    0.9000
         0         0         0    1.0000
```

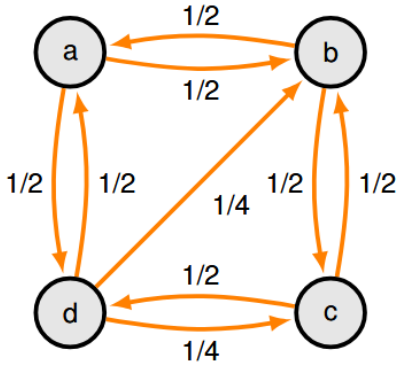
```
P^3 =
    0.1000         0         0    0.9000
         0    0.1000         0    0.9000
         0         0    0.1000    0.9000
         0         0         0    1.0000
```

```
P^4 =
         0         0    0.1000    0.9000
    0.0250         0         0    0.9750
         0    0.0400         0    0.9600
         0         0         0    1.0000
```

```
P^8 =
         0    0.0040         0    0.9960
         0         0    0.0025    0.9975
    0.0010         0         0    0.9990
         0         0         0    1.0000
```

```
P^32 =
         0         0         0    1.0000
         0         0         0    1.0000
         0         0         0    1.0000
         0         0         0    1.0000
```

Example: aperiodic




```
P =
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000    0.2500    0.2500         0
```

```
P^0 =
     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1
```

```
P^1 =
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000    0.2500    0.2500         0
```

```
P^2 =
    0.5000    0.1250    0.3750         0
         0    0.5000         0    0.5000
    0.5000    0.1250    0.3750         0
    0.1250    0.3750    0.1250    0.3750
```

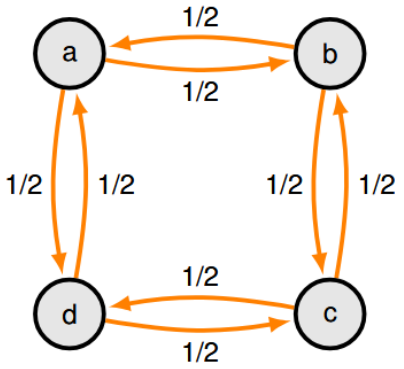
```
P^3 =
    0.0625    0.4375    0.0625    0.4375
    0.5000    0.1250    0.3750         0
    0.0625    0.4375    0.0625    0.4375
    0.3750    0.2188    0.2812    0.1250
```

```
P^4 =
    0.4375    0.1719    0.3281    0.0625
    0.0625    0.4375    0.0625    0.4375
    0.4375    0.1719    0.3281    0.0625
    0.1719    0.3594    0.1406    0.3281
```

```
P^8 =
    0.3564    0.2292    0.2708    0.1436
    0.1572    0.3701    0.1299    0.3428
    0.3564    0.2292    0.2708    0.1436
    0.2156    0.3289    0.1711    0.2844
```

```
P^32 =
    0.2668    0.2927    0.2073    0.2332
    0.2623    0.2958    0.2042    0.2377
    0.2668    0.2927    0.2073    0.2332
    0.2636    0.2949    0.2051    0.2364
```

Example: periodic




```
P =
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
```

```
P^0 =
     1     0     0     0
     0     1     0     0
     0     0     1     0
     0     0     0     1
```

```
P^1 =
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
```

```
P^2 =
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
```

```
P^3 =
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
```

```
P^4 =
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
```

```
P^5 =
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
```

```
P^6 =
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
    0.5000         0    0.5000         0
         0    0.5000         0    0.5000
```

Stationary Distribution

- A distribution $\mu(x)$ on the RV/states such that at time *n*+1 it is the same as at time *n*, is called a **stationary distribution**
  - So called because if the initial state is drawn according to a stationary distribution, the Markov chain forms a stationary SP
- Mathematically, it is the left eigenvector with eigenvalue 1

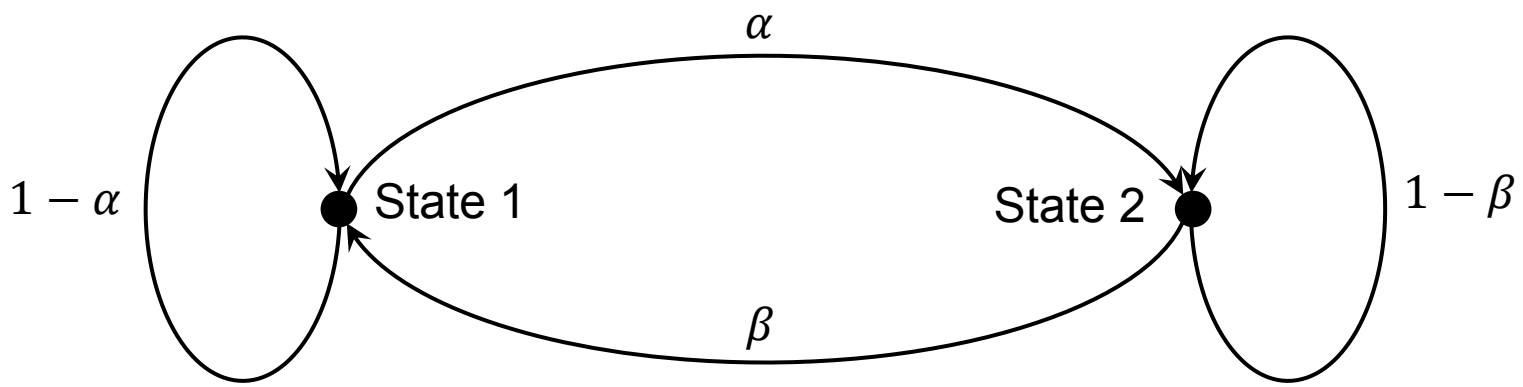
$$\boldsymbol{\mu P = \mu}$$

- If the chain is <u>irreducible</u> and <u>aperiodic</u>, the stationary distribution exists and is <u>unique</u>
- From any starting distribution, the distribution of X_n tends to the stationary distribution as $n \to \infty$

Example: 2-State Markov Chain

- Consider a two-state Markov chain (Bernoulli RV) with a stochastic matrix

$$\mathbf{P} = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix}$$



- The stationary distribution $\boldsymbol{\mu} = [\mu_1\ \mu_2]$ holds probabilities of states 1 and 2 (alphabet of X is $\mathcal{X} = \{\text{state1}, \text{state2}\}$)
- Then the stationary probability can be found by solving the equation $\boldsymbol{\mu}\mathbf{P} = \boldsymbol{\mu}$

$$[\mu_1\ \ \mu_2]\begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} = [\mu_1\ \ \mu_2] \qquad \begin{cases} \mu_1(1-\alpha) + \mu_2\beta = \mu_1 \\ \mu_1\alpha + \mu_2(1-\beta) = \mu_2 \end{cases} \qquad \mu_2 = 1 - \mu_1$$

Example

$\mu_1(1-\alpha)+(1-\mu_1)\beta=\mu_1$ $\mu_1-\alpha\mu_1+\beta-\beta\mu_1=\mu_1$ $\beta=\mu_1(\alpha+\beta)$

- We obtain the stationary distribution

$$\mu_1=\frac{\beta}{\alpha+\beta} \qquad \mu_2=\frac{\alpha}{\alpha+\beta}$$

- If the Markov chain has an initial state drawn according to the stationary distribution, the resulting SP will be stationary
- The entropy of the state X_n at time n is

$$H(X_n)=H_{\mathrm{B}}(\mu_2)=H_{\mathrm{B}}(1-\mu_1)=H_{\mathrm{B}}\left(\frac{\alpha}{\alpha+\beta}\right)$$

- But this is not the rate at which entropy grows for $H(X_1,X_2,\ldots,X_n)$
  - The dependence among the X_i's has a strong impact

Entropy Rate

- We define the **entropy (growth) rate** as the rate at which the entropy of the sequence grows with *n*
- The **entropy (rate)** of a SP $\{X_i\}$ is defined by (when the limit exists)

$$H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X_1, X_2, \ldots, X_n)$$

- We can also define a related quantity for entropy rate (if exists)

$$H'(\mathcal{X}) = \lim_{n\to\infty} H(X_n | X_{n-1}, X_{n-2}, \ldots, X_1)$$

- $H(\mathcal{X})$ and $H'(\mathcal{X})$ are two different notions of entropy rate
  - The 1st is the per-symbol entropy of the *n* RVs
  - The 2nd is the conditional entropy of the last RV given the past
- For stationary SPs both limits exist and are equal

Example: Typewriter

- Consider a typewriter that has m equally likely output letters
- It produces m^n sequences of length n, all of them equally likely
- Hence $H(X_1,X_2,\ldots,X_n) = \log(m^n)$ and the entropy rate is

$$H(\mathcal{X}) = \log m \quad \text{[bits per symbol]}$$



...
'AGCBHIEDF'
'AGCBHIDFE'
'AGCBHIDEF'
'AGCBHFIED'
'AGCBHFIDE'
'AGCBHFEID'
'AGCBHFEDI'
'AGCBHFDIE'
'AGCBHFDEI'
'AGCBHEIFD'
'AGCBHEIDF'
'AGCBHEFID'
'AGCBHEFDI'
'AGCBHEDIF'
'AGCBHEDFI'
'AGCBHDIFE'
...

Example: i.i.d. RVs and Random Walk

- $X_1, X_2, \ldots$ are i.i.d. RVs. Then

$$H(\mathcal{X}) = \lim_{n\to\infty} \frac{H(X_1, X_2, \ldots, X_n)}{n} = \lim_{n\to\infty} \frac{nH(X_1)}{n} = H(X_1)$$

which is what one would expect for the entropy rate per symbol

- For the simple random walk

$$p(x_n|x_{n-1}, x_{n-2}, \ldots, x_1) = p(x_n|x_{n-1}) = \begin{cases} \frac{1}{2}, & x_n = x_{n-1} - 1 \\ \frac{1}{2}, & x_n = x_{n-1} + 1 \end{cases}$$

so

$$H'(\mathcal{X}) = \lim_{n\to\infty} H(X_n|X_{n-1}) = H_B\left(\frac{1}{2}\right) = 1$$

  - Even if the alphabet size grows linearly with *n*, entropy stays unchanged

Theorems for Entropy Rate

- **Theorem:** For a stationary SP, $H(X_n|X_{n-1},\ldots,X_1)$ is nonincreasing in n and has a limit H'($\mathcal{X}$)
  - *Proof:*

$$H(X_{n+1}|X_1, X_2, \ldots, X_n) \leq H(X_{n+1}|X_n, \ldots, X_2) = H(X_n|X_{n-1}, \ldots, X_1)$$

    where the inequality is due to "conditioning reduces entropy" and the equality follows from the stationarity
- Since $H(X_n|X_{n-1},\ldots,X_1)$ is a decreasing sequence of nonnegative numbers, it has a limit, H'($\mathcal{X}$)

- **Th. of Cesàro mean:** Given $a_n \to a$ and the sample mean $b_n = (1/n)\sum_{i=1}^{n} a_i$, then the sample mean $b_n \to a$ too
  - *Informal proof:* Since most of the terms in the sequence $\{a_k\}$ are eventually close to a, then b_n, which is the average of the first n terms, is also eventually close to a

Theorems for Entropy Rate

- **Theorem:** For a stationary SP, the two entropy rate limits exist and are equal

$$H(\mathcal{X}) = H'(\mathcal{X})$$

  - *Proof:* By the chain rule

$$\frac{H(X_1, X_2, \dots, X_n)}{n} = \frac{1}{n}\sum_{i=1}^{n} H(X_i|X_{i-1}, \dots, X_1)$$

    the entropy rate is the time average of the conditional entropies
  - But the conditional entropies tend to a limit *H'*
  - Hence, by Cesàro, their sample mean has a limit, which is equal to the limit *H* of the terms

$$H(\mathcal{X}) = \lim_{n\to\infty} \underbrace{\frac{H(X_1, X_2, \dots, X_n)}{n}}_{b_n} = \lim_{n\to\infty} \frac{1}{n}\sum_{i=1}^{n} \overbrace{H(X_i|X_{i-1}, \dots, X_1)}^{a_i}$$

$$= \lim_{n\to\infty} \underbrace{H(X_n|X_{n-1}, \dots, X_1)}_{a_n} = \overbrace{H'(\mathcal{X})}^{a}$$

Significance of Entropy Rate of SP

- The significance of the entropy rate arises from the AEP
- With probability 1

$$-\frac{1}{n}\log P(X_1, X_2, \ldots, X_n) \xrightarrow{\text{A.S.}} H(\mathcal{X})$$

- The AEP theorems can be extended to a general stationary ergodic process
- We can show that the typical set has a probability close to 1 and that there are about $2^{nH(\mathcal{X})}$ typical sequences of length n, each with probability about $2^{-nH(\mathcal{X})}$
- We can represent the typical sequences using approximately $nH(\mathcal{X})$ bits
- This shows the significance of the entropy rate as the average description length for a stationary ergodic SP

Entropy Rate of Markov Chains

- The entropy rate is well defined for all stationary SP
- For a stationary Markov chain, the entropy rate is given by

$$H(\mathcal{X}) = H'(\mathcal{X}) = \lim_{n\to\infty} H(X_n|X_{n-1}, \dots, X_1) = \lim_{n\to\infty} H(X_n|X_{n-1}) = H(X_2|X_1)$$

where the conditional entropy must be calculated using the given stationary distribution

- The stationary distribution **μ** is the solution of **μP** = **μ**

$$\mu_j = \sum_i \mu_i P_{ij} \qquad \text{for all } j$$

Conditional Entropy

- **Theorem:** Let $\{X_i\}$ be a stationary Markov chain with stationary PMF $\boldsymbol{\mu}$ and transition matrix $\mathbf{P}$. Let $X_1, X_2 \sim \boldsymbol{\mu}$. The entropy rate is

$$H(\mathcal{X}) = \sum_{i=1}^{|\mathcal{X}|} \mu_i \left(-\sum_{j=1}^{|\mathcal{X}|} P_{ij} \log P_{ij} \right)$$

Probability of *i*th row of **P**

Entropy of *i*th row of **P**

  - *Proof:*

$$H(\mathcal{X}) = H(X_2|X_1) = -\sum_{x_1 \in \mathcal{X}} \sum_{x_2 \in \mathcal{X}} p(x_1, x_2) \log p(x_2|x_1)$$

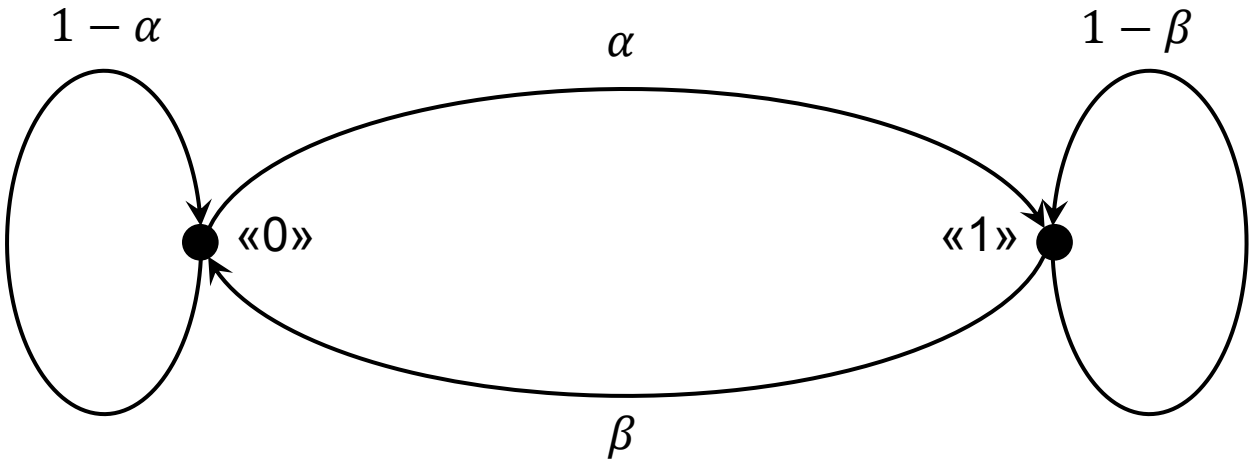
$$= -\sum_{x_1 \in \mathcal{X}} \sum_{x_2 \in \mathcal{X}} p(x_1) p(x_2|x_1) \log p(x_2|x_1) = -\sum_{i=1}^{|\mathcal{X}|} \mu_i \sum_{j=1}^{|\mathcal{X}|} P_{ij} \log P_{ij}$$

- If the Markov chain is irreducible and aperiodic, any initial distribution tends to the unique stationary distribution as $n \to \infty$
- Even though the initial distribution is not the stationary one, the entropy rate, which is a long-term behavior, is $H(\mathcal{X})$

Example: 2-state Markov Chain

- The entropy rate of the two state Markov chain is then

$$H(\mathcal{X}) = H(X_2|X_1) = \frac{\beta}{\alpha+\beta} H_{\mathrm{B}}(\alpha) + \frac{\alpha}{\alpha+\beta} H_{\mathrm{B}}(\beta)$$



$$\mathbf{P} = \begin{bmatrix} 1-\alpha & \alpha \\ \beta & 1-\beta \end{bmatrix} \qquad \mu_1 = \frac{\beta}{\alpha+\beta} \qquad \mu_2 = \frac{\alpha}{\alpha+\beta}$$

Second Law of Thermodynamics

- In statistical thermodynamics, entropy is often defined as the log of the number of microstates in the system
  - It corresponds to our notion of entropy if all the states are equally likely

"**Second Law:** *There exists for every thermodynamic system in equilibrium an extensive scalar property called the entropy, S, such that in an infinitesimal reversible change of state of the system,*

$$dS = dQ/T,$$

where T is the absolute temperature and dQ is the amount of heat received by the system.

The entropy of a thermally insulated system cannot decrease and is constant if and only if all processes are reversible."

Z. S. Spakovszky

Second Law of Thermodynamics

- There can be 5 different interpretations of the 2nd law
  - Let μ_n and μ'_n be two PMFs on the state space of a Markov chain at time n, and μ the PMF of stationary distribution

1. Relative entropy $D(\mu_n \| \mu'_n)$ decreases with time n

Country A and B wealth distributions μ with same taxation system tend to become similar

2. Relative entropy $D(\mu_n \| \mu)$ between a distribution and the stationary distribution decreases with time n

Wealth distribution tends to a stationary limit μ

3. Entropy $H(X_n)$ increases with n if the stationary distribution is uniform

Uniform starting PMF & nonuniform stationary PMF -> H decreases

Uniform stationary PMF

$$D(\mu_n \parallel \mu) = \log|\mathcal{X}| - H(\mu_n) = \log|\mathcal{X}| - H(X_n)$$

4. The conditional entropy $H(X_n|X_1)$ increases with time n for a stationary Markov chain

Conditional uncertainty on the future increases with time

5. The conditional entropy $H(X_0|X_n)$ of the initial condition X_0 increases for any Markov chain

Past becomes uncertain after a while

Functions of Markov Chains

- Let $X_1, X_2, \ldots, X_n, \ldots$ be a stationary Markov chain, and let $Y_i = \varphi(X_i)$ be a SP function of the corresponding state in the Markov chain
- This occurs in practice, but what is the entropy rate $H(\mathcal{Y})$?
- It would simplify matters greatly if $Y_1, Y_2, \ldots, Y_n$ also formed a Markov chain, but in many cases, this is not true
- Since the Markov chain is stationary, so is $Y_1, Y_2, \ldots, Y_n$, and the entropy rate is well defined
- However, if we wish to compute $H(\mathcal{Y})$, we might compute $H(Y_n \mid Y_{n-1}, \ldots, Y_1)$ for each n and find the limit
- Since the convergence can be arbitrarily slow, we will never know how close we are to the limit

Functions of Markov Chains

- We already know that $H(Y_n|Y_{n-1},\dots,Y_1)$ converges monotonically to $H(\mathcal{Y})$ from above
- For a lower bound, we will use $H(Y_n|Y_{n-1},\dots,Y_1,X_1)$
- This is a neat trick based on the idea that X_1 contains as much information about Y_n as Y_1,Y_0,Y_{-1}, …

- **Theorem:** If $X_1,X_2,\dots,X_n$ form a stationary Markov chain, and $Y_i = \varphi(X_i)$, then

$$H(Y_n|Y_{n-1},\dots,Y_1,X_1) \le H(\mathcal{Y}) \le H(Y_n|Y_{n-1},\dots,Y_1)$$

and

$$\lim_{n\to\infty} H(Y_n|Y_{n-1},\dots,Y_1,X_1) = H(\mathcal{Y}) = \lim_{n\to\infty} H(Y_n|Y_{n-1},\dots,Y_1)$$

  - *Proof:* this is based on the Lemmas

$$H(Y_n|Y_{n-1},\dots,Y_2,X_1) \le H(\mathcal{Y})$$

and

$$H(Y_n|Y_{n-1},\dots,Y_1) - H(Y_n|Y_{n-1},\dots,Y_1,X_1) \to 0$$

Hidden Markov Models

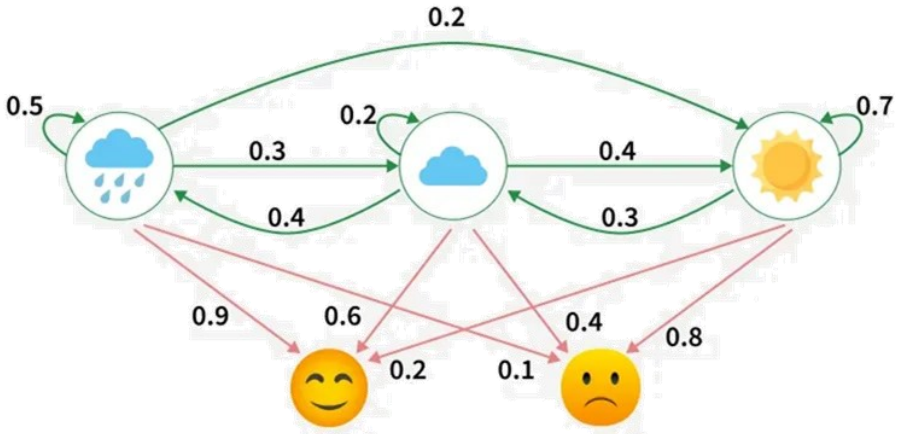
- In general, we could also consider the case where Y_i is a stochastic function (nondeterministic function) of X_i
- Consider a Markov process $X_1, X_2, \ldots, X_n$, and define a new process $Y_1, Y_2, \ldots, Y_n$, where each Y_i is drawn according to $p(y_i|x_i)$, conditionally independent of all the other X_j, $j \neq i$

$$p(x^n, y^n) = p(x_1) \prod_{i=1}^{n-1} p(x_{i+1}|x_i) \prod_{i=1}^{n} p(y_i|x_i)$$

- Such a process, called a **hidden Markov model (HMM)**, is used extensively in speech recognition, handwriting recognition, etc.
- The same argument as that used above for functions of a Markov chain carry over to HMMs, and we can lower bound the entropy rate of an HMM by conditioning it on the underlying Markov state

Example of HMM

- The actual system state is hidden
  - Weather (Sunny/Cloudy/Rainy)
- Outcomes by the hidden states are observed
  - Friend mood (Happy/Sad)



- Transition matrix: stochastic matrix of hidden states (green)
- Emission matrix: probability of observing an outcome given a hidden state (red)

1. How to compute the probability of an observation sequence? (*Forward algorithm*)
2. How to find the most likely hidden state sequence that explains the observations? (*Viterbi algorithm*)
3. How to train the HMM to fit the observed data? (Baum-Welch *EM algorithm*)